

Shuowen Li

Department of Precision Instrument, Tsinghua University

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Education

Tsinghua University, M.S. in Optical Engineering Sep. 2023 – Jun. 2026

- Advisor: Prof. Liangcai Cao
- Overall GPA: 3.64/4.0

Lanzhou University, B.S. in Physics Sep. 2019 – Jun. 2023

- Advisor: Prof. Hao Jia
- Ranking: 1/291; Overall GPA: 4.23/5.0
- Awarded the **National Scholarship** in 2020

Research Experience

Human-AI Interaction and Creative Robotics Sep. 2023 – Present

Academy of Arts & Design, Tsinghua University (Advisor: Prof. Haipeng Mi)

- [Developed a semantic-guided generative tool for artificial life design](#), enabling designers to evolve digital lifeforms via natural language prompts. Integrated CLIP-based evaluation and CMA-ES optimization in a real-time simulation engine to support interactive evolution. A conference paper is under submission.
- [Engineered a swarm robotic storytelling platform \(PuppetLine\)](#) for children's creative expression using Toio robots. Built a natural language-driven robot control system using large language models and a library of action and emotion primitives. The system maps users' narrative input into multi-robot scripts, enabling a novel form of co-creative storytelling. A conference paper is under submission.
- [Conceived and developed a wearable puppet robot \(Embo\)](#) that translates verbal aggression into tactile pressure feedback. Designed for child-robot interaction in empathy education, the system transforms language severity into sensory discomfort, offering emotionally responsive intervention for bullying scenarios. Published a related conference paper.

Computational Imaging and 3D Display System Sep. 2023 – Present

Department of Precision Instrument, Tsinghua University (Advisor: Prof. Liangcai Cao)

- [Designed and evaluated a hardware and metric framework for autostereoscopic \(glasses-free 3D\) displays](#), focusing on human visual comfort, crosstalk levels, and depth perception. Integrated computational imaging models with perceptual evaluation methods, contributing to the development of industry standards for autostereoscopic displays. Participated in drafting a group standard and published a related research paper.
- [Developed real-time reconstruction algorithms and a four-aperture telescope array for turbulence-degraded astronomical imaging](#), enabling high-fidelity atmospheric turbulence removal. Improved and integrated blind deconvolution, deformable convolution, and diffusion model. Filed a patent and published a research paper.
- [Conducted a survey on lensless computational imaging](#), covering optical encoding techniques, image sensor integration, and algorithmic reconstruction strategies. Outlined potential applications in compact and modular imaging systems across scientific and industrial domains. Published a related review paper.

Publications

Academic Papers ([†] co-first author)

1. **Shuowen Li[†]**, Kexin Wang[†], Minglu Fang, Danqi Huang, Ali Asadipour, Haipeng Mi, and Yitong Sun. “[Participatory Evolution of Artificial Life Systems via Semantic Feedback](#)”. In: *SIGGRAPH Asia 2025 Art Papers*.
2. Ruhan Wang, **Shuowen Li**, Peiran Zhang, Danqi Huang, Yijie Guo, and Haipeng Mi. “[PuppetLine: An Interactive System for Embodied Storytelling with LLM-driven Swarm Robots](#)”. In: *Adjunct Proceedings of the*

38th Annual ACM Symposium on User Interface Software and Technology (UIST), 2025.

3. Shihan Qiu[†], Yuhao Xie[†], **Shuowen Li**[†], Wei Guo, Xiaoyue Gao, and Yijie Guo. “Embo: A Wearable Robot Transforming Child-Directed Verbal Aggression into Tactile Feedback”. In: *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)* (2024), pp. 857–861.
4. **Shuowen Li** and Liangcai Cao. “Multidimensional Crosstalk Analysis in Autostereoscopic Displays: Integrating Subjective and Objective Evaluations for Image Quality Assessment”. In: *Optics Express* **33**(8), 2025, pp. 16911–16924. (*JCR Q2, IF 3.3*).
5. **Shuowen Li**, Yunhui Gao, Jiachen Wu, Mingjie Wang, Zhangcheng Huang, Shumei Chen, and Liangcai Cao. “Lensless Camera: Unraveling the Breakthroughs and Prospects”. In: *Fundamental Research* **5**(4), 2025, pp. 1725–1736. (*JCR Q1, IF 6.3*).
6. **Shuowen Li**, Yunhui Gao, Jiachen Wu, and Liangcai Cao. “Blind Deblurring of Astronomical Images Using SCGTV-Based Single-Frame Method”. In: *Optics Express* **32**(20), 2024, pp. 35579–35593. (*JCR Q2, IF 3.3*).

Book Translation

1. Geoff Cottrell. “Matter: A Very Short Introduction”. Translated by Xiang Liu, **Shuowen Li**, and Jiageng Li. Yilin Press (2024). ISBN: 9787575301671.

Patents

1. Liangcai Cao, **Shuowen Li**, Jiachen Wu. “Single-frame astronomical image reconstruction method, device, electronic equipment and storage medium”. Chinese Patent CN202411139297X. Under substantive examination.

Conference Presentations

1. **Shuowen Li** and Liangcai Cao. “Image quality evaluation methods and systems for three-dimensional displays”. In: *13th National Conference on Photonics*, Lanzhou, China (2024). Outstanding Oral Presentation.
2. **Shuowen Li** and Liangcai Cao. “Blind deblurring of astronomical image using SCGTV-based single-frame method”. In: *SPIE/COS Photonics Asia*, Nantong, China (2024). Oral Presentation.
3. Yunhui Gao, **Shuowen Li** and Liangcai Cao. “Lensless Computational Imaging Framework with Applications to Digital Holography”. In: *OPIC OPTICS & PHOTONICS International Congress*, Japan (2024). Poster Presentation.

Honors and Awards

- Tsinghua Friends–Xiaogan Talent Scholarship (2024)
- Outstanding Graduate of Lanzhou University (2023)
- Outstanding Student of Gansu Province Universities (2021)
- National Scholarship (2020)
- First-Class Scholarship of Lanzhou University (2020–2023) (*Awarded consecutively for three years*)

Skills and Interests

- **Programming:** Python, MATLAB, Java, C++
- **Softwares:** Blender, Unity, SolidWorks, Arduino, Adobe Illustrator, Lumerical FDTD, Zemax
- **Interests:** Amateur photographer, Film enthusiast, Art exhibitions, Sports

Referee

- Prof. Haipeng Mi (Academy of Arts & Design, Tsinghua University)
Email: mhp@tsinghua.edu.cn
- Prof. Hao Jia (School of Physical Science and Technology, Lanzhou University)
Email: jiahao@lzu.edu.cn